Ask for JSON: Structured Output Collapses Answer Diversity Across 44 Language Models

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Abstract

When a language model must choose one answer from a large space of equally valid options, a format clause — *"Reply with JSON only"* — changes which answer it chooses. We re-run the One-Word Census [1], 31 category prompts with wide answer spaces asked of 44 models, with the reply requested in JSON: no schema enforcement, no constrained decoding, only the request. The field's convergence deepens sharply: on the unconstrained prompt ("Pick a word."), the modal answer's share rises from 40% to 63% and distinct answers fall from 53 to 28; mean answer-choice surprisal drops 1.80 to 1.58 bits. The tax is *progressive* and respects the instrument's resolution: seven of 44 models move individually (BH-FDR $q = .10$), all toward the mode, led by the panel's two most distinctive frontier models (the strongest explorer loses 1.32 bits — half its measured distinctiveness) while the conformist floor is unmoved; one family-level effect survives aggregation — the Meta-tuned lineage *diverges* under JSON (+0.70 bits against the rest of the field, $p < 10^{-4}$) — and the remaining deltas form a noise plateau we decline to rank. The format acts as a sharpener, not a re-indexer — the plain-chat modal answer survives in 28 of 31 categories, and the three flips occur exactly where the plain mode was weakest. The column also dissociates two kinds of distinctiveness the plain census conflates, read from discrete behavior rather than score deltas: GPT-5.6 Terra's stable off-modal defaults (*mustard*, *mango*, four runs of four in chat) vanish inside the register (0% modal avoidance in XML), while Claude Fable 5 holds the same defaults four-of-four — divergence as costume vs. conviction. Full-battery controls reveal a register gradient: the compression is significant and specific to the answer-delivery formats models are trained to speak (JSON −0.22 bits, $p = 10^{-4}$; XML −0.18, $p = .003$; sign-flip permutation over models, corroborated over categories), absent for YAML and CSV ($p > .15$), and *reversed* for an arbitrary non-data wrapper — square brackets significantly loosen the field (+0.16, $p = .002$). On the unconstrained prompt, all four serializations concentrate the pool (serendipity 41% plain → 52–65%), so a corpus-register component remains visible; but the battery-wide gradient weighs toward tool-use post-training as the dominant mechanism. Structured output is how software consumes language models; that surface is served by a measurably more homogeneous model than the chat surface on which models are evaluated, compared, and chosen.

1 Introduction

2 Related work

3 Method

3.1 Instrument

3.2 Format columns

3.3 Parsing and hygiene

4 Results

4.1 The format tax is progressive

4.2 A sharpener, not a re-indexer

4.3 The register moves the mode, not the temperature

4.4 Conviction vs. costume

4.5 Serialization vs. structure: the register gradient

5 Limitations

6 Discussion

Data and code availability

Note on AI usage

References

[1] Tapan Parikh. The one-word census: Answer-choice conformity across 39 language models, 2026. arXiv preprint; data and explorer at https://github.com/tap2k/modelun.